

Perovskite Photovoltaics in July 2026: Trap density and ion migration, self-assembled monolayer contacts

Havid Aqoma^{1,2,3,4,*}

¹School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

²Kelip-kelip! Center of Excellence for Light Enabling Technologies, School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

³Centre of Excellence for Green and Advanced Technologies in Efficient Separations (GATES), School of Energy and Chemical Engineering, Xiamen University Malaysia, Selangor Darul Ehsan 43900, Malaysia

⁴Gulei Innovation Institute, Xiamen University, Zhangzhou 363200, China

*Corresponding author. Email: havidaqoma.khoiruddin@xmu.edu.my

September 9, 2026

Abstract

Across 648 indexed publications evaluated this month, with 181 examined in depth, the verified performance ceiling reaches a power conversion efficiency (PCE) of 33.1% in monolithic tandems and 27.31% in single-junction inverted devices. Tandem benchmarks rely on current matching across complementary bandgaps, whereas single-junction records depend on multidentate chelating agents and self-assembled monolayer (SAM) contacts that passivate undercoordinated lead cations and align interfacial dipoles. Yet only 22 closely read studies provide independently certified efficiency values, as uncertified reports frequently mask mobile ionic defects and non-radiative recombination under transient voltage sweeps. Scaling introduces severe resistive and geometric transport losses, establishing an area penalty that restricts certified efficiency to 27.31% at the largest reported aperture area of 766 cm². This scaling deficit arises because molecular passivators that eliminate trap states on microscale cells aggregate during meniscus drying, intensifying non-radiative recombination across laser-scribed module interconnects. Operational durability evidence remains narrow across the literature: only 4 investigations quantify lifetime metrics, with the longest duration reaching 9800 h, while only 6 studies adhere to standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols. A pronounced reporting disparity persists between claimed and verified performance, with 50.4% of abstracts reporting efficiency but only 8.3% documenting independent certification. Mechanistic evaluation demonstrates that contact stabilization through self-assembled monolayers (SAMs) and multidentate coordination fails to establish resilience against thermo-mechanical fatigue, as thermal expansion mismatches induce cyclic shear stresses that trigger interfacial delamination. Depth-resolved degradation profiles confirm that breakdown initiates at buried selective contacts before penetrating bulk absorbers. Chemical treatments and interlayers also fail to achieve permanent suppression of halide phase segregation in wide-bandgap absorbers, where illumination and thermal cycling drive phase demixing without resolving the boundary between thermodynamic stability and kinetic demixing arrest. Finally, unstandardized aging without continuous tracking leaves the division between reversible ionic polarization and irreversible lattice degradation unquantified.

Keywords: perovskite photovoltaics; buried interface; defect passivation; wide-bandgap perovskite;

tandem solar cells; operational stability

1. Introduction

Perovskite photovoltaics has reached a juncture where incremental gains in laboratory power conversion efficiency (PCE) no longer mark material progress. The defining frontier has shifted to independent verification, scalability across module-relevant area, and operating lifetime under operational conditions. Peak metrics obtained from uncertified, sub-millimetre devices under transient voltage sweeps frequently obscure mobile ionic defects and non-radiative interfacial recombination that manifest only over extended timescales.

Within this landscape, 648 indexed works met the scope gate in July 2026 and 181 were read closely.

The mechanistic threads followed in this issue encompass passivation chemistry and ion migration under bias, self-assembled contacts and interfacial dipoles, precursor chemistry and phase purity, stability under load, heat and illumination, and wide-bandgap subcells and recombination layers. Together, these processes govern whether energetic band alignment and chemical defect suppression survive the coupled electrochemical gradients of field operation.

To evaluate these developments, this review distinguishes what the month’s literature verified from what it asserted. Filtering out artifacts of fast scan rates, uncalibrated spectral illumination, and negligible active areas isolates the specific chemical and structural interventions that provide authentic stability under working conditions.

2. The Certified Frontier in July 2026

The verified performance frontier reaches a power conversion efficiency (PCE) of 33.1% in two-terminal (2T) tandem architectures and 27.31% in single-junction inverted (p-i-n) devices. Yet only 22 of 169 closely read papers reported an independently certified efficiency. In 2T tandems, certified values of 33.1% and 32.13%, along with intermediate benchmarks at 29.57%, 28.84%, and 27.49%, confirm that current matching across paired subcells dictates the verifiable ceiling [1,2,3,4,5]. Across p-i-n single junctions, certified PCE concentrates tightly at 27.31% and 27.3%, establishing an interfacial recombination barrier where non-radiative losses at transport layers suppress open-circuit voltage [6,7,8].

Area expansion reveals how uncertified laboratory reports outpace verified metrics. For a 625 cm² aperture, self-reported efficiency reached 27.8% despite an independently certified baseline of 27.3% [8]. Certified large-area devices quantify the resistive and geometric penalties imposed by macroscopic transport. An 818 cm² device achieved 26.53% [9]. A 766 cm² device sustained 27.31% [7]. A 684.75 cm² device reached 27.31% [6]. A 655.2 cm² device delivered 27.12% [10]. These metrics verify that scalable interconnects and spatial uniformity can suppress series resistance across large modules.

A champion efficiency obtained from an unshadowed, sub-0.1 cm² laboratory cell and an independently certified measurement across a module represent irreconcilable categories of physical evidence. Microscale test structures probe localized carrier generation under negligible series resistance, systematically masking pinhole shunting, film non-uniformity, and perimeter recombination. Conversely, certified module-scale metrics integrate transparent electrode sheet resistance, scribe dead zones, and macroscopic phase segregation across active layers. Equating isolated laboratory spikes with certified module performance mistakes local material potential for verified device technology.

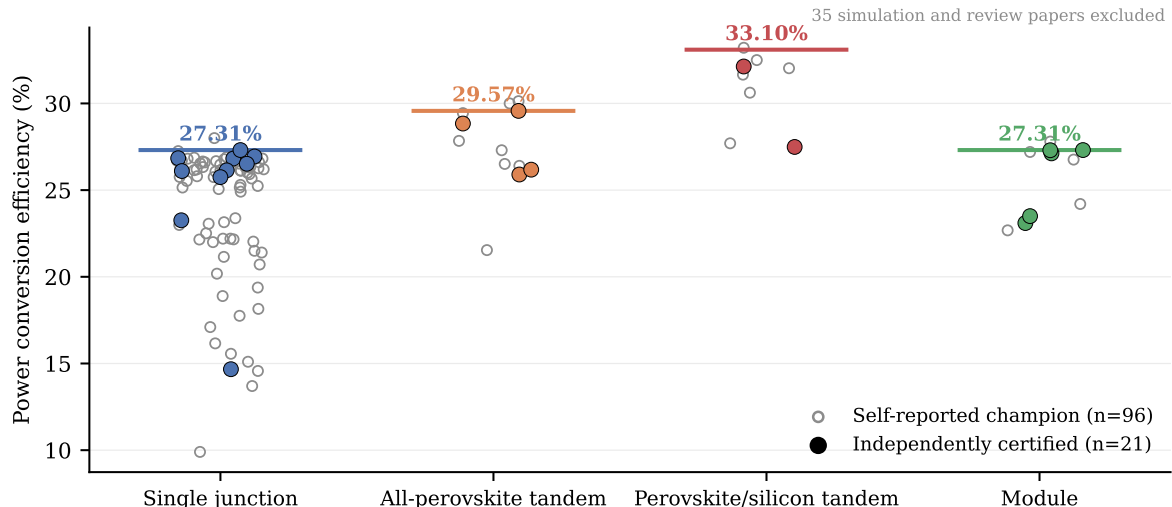


Figure 1. The certified frontier in July 2026. Filled markers are independently certified efficiencies, open markers are self-reported champion values, grouped by device family. Horizontal bars mark the highest certified value in each family.

3. Passivation Chemistry and Ion Migration Under Bias

The quantitative frontier for perovskite solar cells now reaches certified power conversion efficiency (PCE) values of 26.84% (champion 27.25%) in inverted devices and 23.5% on 65.05 cm² modules (champion 27.2%). Uncertified small-area claims reach 26.85% [11,12,13]. These benchmarks force a critical evaluation of what the reported physical evidence actually licenses. One possibility is that interfacial chemical treatments permanently suppress deep trap states and mobile ion channels under operating bias. The alternative is that they merely provide transient electrostatic screening that degrades during continuous operation.

At perovskite interfaces, undercoordinated Pb²⁺ cations and halide vacancies constitute primary non-radiative recombination centers. Monodentate Lewis bases donate electron pairs from sulfur, nitrogen, or oxygen to passivate surface Pb²⁺ defects. Carbazole-based self-assembled monolayers (SAMs) such as BTP at buried interfaces, thiophene sulfur coordination, and carboxyl/sulfur bifunctional TTA demonstrate this route. These three report uncertified champion PCE values of 26.85%, 21.15%, and 26.7% [13,14,15]. However, monodentate coordination offers limited thermal resistance against cooperative defect formation. To address this, amphoteric ionic liquids (GUHS and GUDHP) and FTAC simultaneously passivate Lewis acidic and basic sites while compensating iodine vacancies. GUDHP achieves 20.06% PCE with 85.74% fill factor (FF) at 1.84 eV bandgap (champion 26.13%). FTAC retains 88.67% of initial efficiency after 1,0 h maximum-power-point tracking (maximum power point tracking (MPPT)) at 65 °C despite an uncertified 25.06% champion PCE [16,17]. Multivalent spatial anchoring further impedes defect formation. Three-dimensional distribution of amino groups in tetrakis(4-aminophenyl)methane (4APM), nonlayered intermediate-site anchoring ligands, and m-bis(trifluoromethyl)aniline (mTFMA) bind PbI₂ or formamidinium iodide (FAI) precursors. That binding lengthens carrier lifetimes from 529.7 to 2479.4 ns. These films retain over 95% efficiency after 1000 h MPPT at 65 °C across 14 cm² (certified 26.26%). They also retain over 90% PCE after 2000 h at 65 °C and 90% relative humidity (RH) (uncertified champion 26.54%, projected T80 of 9800 h) [18,19,20].

Under continuous bias, applied electric fields drive mobile iodide migration toward extraction layers, causing interfacial degradation and phase segregation. Chemical immobilization directly arrests this migration. Dual iodide-trapping agents (TFIBA and FPIFA), two-dimensional/three-dimensional (2D/3D) halide imidazolium heterojunctions, and 1-(4'-sulfophenyl)-3-carboxyl-5-pyrazolone sodium salt (SCP) layers suppress spiro-OMeTAD and electron-contact degradation. The first route achieves an uncertified 26.81% champion PCE with T93 lifetime over 1140 h at 85 °C. The second holds 92% retention after 1200 h at 85 °C (champion 26.33%). The third holds 90% retention after 1485 h MPPT (certified 26.84%, champion 27.25%) [11,21,22]. In wide-bandgap absorbers, field-driven halide separation is suppressed by guanidine phosphate (champion 21.54% at 1.78 eV, retaining 90.64% after 1400 h). It is also suppressed by acetylthiocholine chloride (ATC) doping, which fills vacancies with quaternary ammonium and chloride ions to restrain demixing across 63.7 cm² modules (champion 22.68%) [23,24]. Conformal plasma surface engineering similarly mitigates defect accumulation at laser-scribed interconnects to secure a certified 23.5% module PCE on 65.05 cm² (champion 27.2%). Small-area devices retained 98.1% of initial PCE after 2000 h MPPT at 85 °C [12].

Lattice strain fundamentally governs ion migration kinetics by lowering vacancy hopping barriers under tensile expansion. Relieving this strain through dual-site coordination with dithiopyr bridges adjacent Pb²⁺ sites, promoting preferential (100) crystal orientation and yielding a certified 26.14% PCE (champion 26.9%, and 21.28% on 30 × 30 cm² modules) [25]. Compressive strain engineering counteracts tensile distortion and impedes defect formation through three routes. The polymer MPUA reaches a champion 24.91%. Cross-linked 2-amino-3-hydroxybutanoic acid (AHBA) interfaces accelerate electron extraction to 3.5 ps (champion 26.32%). Photoresponsive C₆₀-azo trans-to-cis isomerization under ultraviolet (UV) light gives 92.7% retention after 488 h (champion 26.6%) [26,27,28]. Nevertheless, most strain and passivation studies report uncertified, small-area metrics without operando tracking of mobile ion accumulation under concurrent bias, heat, and illumination, leaving long-term field immunity unsubstantiated.

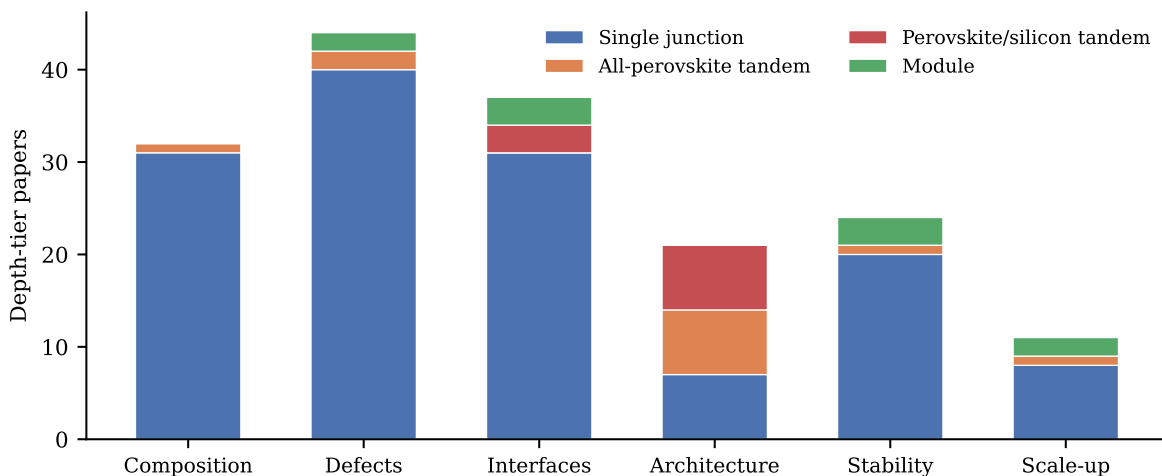


Figure 2. Where the month’s mechanistic effort was directed. Depth-tier papers by mechanism axis, stacked by device family.

4. Self-Assembled Contacts and Interfacial Dipoles

The quantitative frontier for contact-engineered perovskite photovoltaics is established by monolithic tandem configurations achieving certified power conversion efficiency (PCE) values of 33.1% (champion 33.3%) [1] and single-junction inverted architectures reaching certified 27.31% [6]. Translating these efficiencies into durable devices hinges on resolving three core physical problems: weak oxide anchoring, self-assembled monolayer (SAM) aggregation, and interfacial dipole misalignment.

Substrate delamination and thermal desorption originate from fragile monodentate coordination at transparent conducting oxides. Mechanical toughness is markedly substrate-dependent: depositing self-assembled monolayers (SAMs) on a nickel oxide (NiO_x) underlayer increases interfacial fracture energy tenfold relative to indium tin oxide (ITO), while removing loosely bound molecules elevates fracture energy from 0.94 to 4.82 J m^{-2} [29]. Multidentate chelating motifs directly counter this vulnerability by coordinating undercoordinated lead (Pb^{2+}) and reinforcing substrate anchoring. Tetradentate iron(III) porphyrins yield certified 26.82% PCE (champion 27.06%) [30]. Bisphosphonates reconstruct the NiO_x /SAM buried interface to reach certified 27.31% PCE on small-area rigid cells [6]. Zoledronic acid (ZDA) head groups yield 30.62% tandem PCE, and the wide-bandgap sub-cells (20.55% PCE) retain over 90% performance after 200 h of maximum power point (MPP) tracking [6,30,31]. Lindqvist-type polyoxometalates similarly raise the anchoring ratio on ITO from 29.93% to 53.98% and cut the desorption rate from 30.7% to 5.6% during continuous MPP tracking (champion 25.15% PCE) [32]. Large-area module demonstrations remain scarce, though tetraphosphorylated phthalocyanines demonstrate a certified 23.1% PCE (stabilised 22.9%) on a 20.3 cm^2 module without degradation after 61 days of International Summit on Organic Photovoltaic Stability (ISOS)-O-1 outdoor aging [33].

Intermolecular pi-pi stacking among aromatic cores drives nanoscale SAM agglomeration, generating pinholes and phase inhomogeneities. Disrupting these interactions via steric hindrance or electrostatic confinement suppresses clustering. Noncoplanar multidentate contacts (TTA) impede pi-pi stacking to stabilize wide-bandgap single-junction absorbers (approximately 1.67 eV, champion PCE of 23.7%) and tandems [1]. Ionic liquid lubricants (BMIMBF₄) suppress aggregation via dual steric-electrostatic modulation. Those devices retained 92.3% of initial efficiency after 800 h at 85 °C under air mass 1.5 global (AM 1.5G) illumination (uncertified champion 26.82% PCE) [34]. Concurrently, proton transfer complex-confined carbazole SAMs maintain thermal stability at 75 °C for 1368 h (champion 26.23% PCE) [35], and in situ thermally activated polymerization networks seal interfacial voids to achieve 26.87% champion PCE [34,35,36].

Unfavorable vacuum-level shifts and unpassivated defects dictate carrier extraction deficits at the absorber boundary. Halogenation and co-assembly modulate work functions. Substituting methoxy groups with fluorine in F-2PACz strengthens the interfacial dipole and passivates halide vacancies. That contact retains 25.06% PCE on 1 cm^2 devices with 3.11% relative efficiency loss (champion 25.75%) [37]. Co-assembling crown ethers at electron transport layer interfaces instead increases surface coverage from 0.57 to 0.79 and strengthens interfacial dipole moments from 2 to 7 Debye (champion 26.33% PCE) [38]. Polycyclic perylenecarbazole cores (6PCzBr) tailor electrostatic potentials for 26.20% PCE [39]. Phosphine additives (TFMP) coordinate Pb^{2+} and stabilize formamidinium (FA^+) in Me-4PACz and 4PABCz ternary contacts (champion 26.78% PCE) [40]. Ammonium bromide interlayers (PFN-Br) tune band alignment for 23.15% PCE [41]. However, because most dipolar gains are reported on uncertified sub- 0.1 cm^2 cells without standardized bias protocols, whether these electrostatic barriers resist operational ion drift remains unverified.

5. Precursor Chemistry and Phase Purity

The quantitative frontier for perovskite photovoltaics is defined by single-junction power conversion efficiency (PCE) reaching certified values of 26.52% (26.88% champion) [42] and uncertified reports exceeding 26% [43,44]. Whether these metrics reflect phase-pure lattices or metastable polymorphic mixtures remains governed by precursor kinetics, phase segregation, and spatial scalability. Record efficiencies frequently rely on sub-0.1 cm² apertures, such as 26.32% on 0.072 cm² [43]. Scaling to a 900 cm² module lowers efficiency to 20.25% (17.75% certified). That module still retained over 91% initial efficiency after 1000 hours at 85°C and 85% relative humidity (RH) [42].

Controlling precursor coordination prevents solvent degradation and defective phase precipitation [45,46,47]. In formamidinium (FA) lead triiodide (FAPbI₃) growth, parasitic esterification between gamma-butyrolactone (GBL) and 2-methoxyethanol (2ME) destabilizes heated solutions; maintaining lead iodide (PbI₂) deficiency delays redissolution to suppress delta-phase impurities, yielding 22.99% PCE [45]. Additives also eliminate antisolvent quenching: sulfolane forms an intermediate phase yielding 22.2% PCE in methylammonium (MA) lead triiodide (MAPbI₃) [46], while 3-furoic acid (3-FA) coordinates tin iodide (SnI₂) and hydrogen-bonds with formamidinium iodide (FAI) to decouple nucleation from crystal growth, reaching 16.16% PCE [47]. Stoichiometric imbalance leaves defective binary halides: PbX₂-terminated surfaces display higher defect densities and lower stability than AX terminations [48]. Passivating residual PbI₂ with triphenyl phosphoramidate (TPPA) yields 21.4% PCE and retains 90% initial efficiency after 750 hours of maximum power point tracking (MPPT) versus 270 hours for controls [49]. Similarly, molten 4-(4,6-dimethoxy-1,3,5-triazin-2-yl)-4-methylmorpholinium chloride (DMTMM) dissolves two-dimensional (2D) phases to eliminate interfacial PbI₂, raising PCE from 23.43% to 25.73% [50], while optimizing PbI₂ evaporation distance achieves 14.57% PCE over 64 cm² [51].

Beyond nucleation, compositional heterogeneity drives phase demixing [52,53]. Concurrent four-dimensional scanning transmission electron microscopy (4D-STEM) and energy-dispersive X-ray spectroscopy (EDX) reveals that photoinactive hexagonal polytypes nucleate from nanoscale cesium (Cs) deficiencies [52]. In mixed-halide absorbers, strontium doping prevents I₂Br- intermediate formation, suppressing segregation to reach 20.18% PCE at 1.77 eV and 19.02% at 1.84 eV [54]. Rubidium A-site incorporation induces lattice distortion to suppress halide demixing, providing an open-circuit voltage (Voc) of 1.46 V in wide-bandgap sub-cells and 3.31 V in triple-junction tandems [55]. Conversely, trivalent bismuth substitution in cesium lead halides (CsPbX₃) triggers phase segregation due to lower thermodynamic stability [53]. Dielectric-response screening of 5043 compositions confirms that phase stability requires narrow windows of 2–8% Cs, 78–89% FA, 3–18% MA, and 93–99% iodide (I) [56].

Dimensional heterostructuring stabilizes interfaces, but phase purity remains constrained by transport penalties and area scaling [43,44,57]. In situ hydrogen-bonded hydroxyphenethylammonium (HPA) spacer networks retain 95.5% of initial efficiency after 3000 hours at 30°C in N₂, 92.5% in 45% ± 5% RH air, and 90% after 800 hours at 85°C in N₂, but on 0.072 cm² cells at 26.32% PCE [43]. Colloidal suspension epitaxy yields deterministic 26.0% multilayers without disturbing underlying layers [44]. In flexible tin cells, interfacial hydrogen bonding yields 15.56% PCE (14.67% certified) on 0.049 cm², but drops to 11.06% at 1 cm² despite achieving 22.21% under 1000 lux indoor illumination [57]. While precursor coordination and dimensional stabilization preserve microscopic phase purity, sub-0.1 cm² testbeds obscure severe interfacial penalties encountered during physical scaling.

6. Stability Under Load, Heat and Illumination

The quantitative frontier for operational durability is defined by time to 80% initial performance (T80) values of 9800 h [19] and 6785 h [58], falling steeply to 720 h [59] and 200 h [60]. However, this frontier rests on narrow evidence. Only 4 of 169 closely read studies report a T80 metric. Only 6 identify standardized International Summit on Organic Photovoltaic Stability (ISOS) protocols, such as ISOS-O-1 [33], ISOS-L-1 [7,61], ISOS-L-3 [62], ISOS-L-11 [9], or continuous maximum power point tracking (MPPT) [32]. Most stability claims instead rely on unstandardized, short-duration testing.

Depth-resolved profiling reveals that photo-induced degradation originates at both top and buried interfaces before penetrating the bulk [63]. To arrest interfacial chemical breakdown, molecularly tailored self-assembled monolayers (SAMs) stabilize buried contacts. These contacts yield a certified power conversion efficiency (PCE) of 27.31% in single cells and 21.54% on a 766 cm² module under ISOS-L-1 [7]. Asymmetric SAMs achieve 20.59% on 655.2 cm² modules without certified durability [64]. At the upper contact, multidentate fullerenes passivate defects and immobilize surface cations under ISOS-L-3, retaining 98.7% of initial PCE after 2100 h of continuous illumination and 95.8% after 1800 h at 85 °C, though on uncertified devices [62]. Interfacial defect passivation via aromatic carbonyls similarly claims stability but remains confined to an uncertified, sub-0.1 cm² cell of 0.09 cm² [65]. In-situ two-dimensional perovskite reconstruction further blocks interfacial degradation [66,67], while volatile dopants eliminate mobile lithium-ion residues in hole-transporting layers [68].

Under continuous bias and illumination, trapped-charge accumulation accelerates halide phase segregation, degrading wide-bandgap absorbers under standard MPPT, whereas maximum stability point tracking (MSPT) retains approximately 90% of initial power after 700 h [69]. Ion-migration barriers from diammonium additives suppress segregation to achieve a certified 26.18% PCE on a 1.17 cm² triple-junction aperture [70], aligning with rubidium doping to homogenize vertical cation distributions under ISOS-L-1 [61]. These degradation pathways are intensified by dynamic thermo-mechanical stress fluctuations [71], where inelastic tensile stress pre-relaxation reduces moisture ingress by up to an order of magnitude [72]. Without widespread adoption of standardized ISOS protocols, reported operational lifetimes remain poorly substantiated.

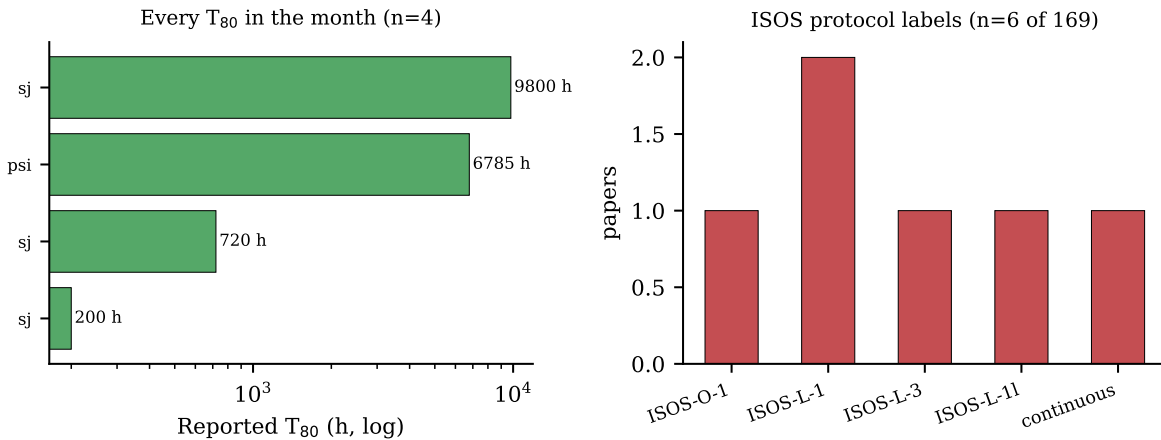


Figure 3. The month’s operational-stability evidence in full. Left, every reported time to 80% of initial performance. Right, the stability protocol labels used.

7. Wide-Bandgap Subcells and Recombination Layers

Monolithic perovskite-silicon tandems have pushed power conversion efficiency (PCE) to 33.21% (certified 32.13%) and 32.5% through interface and crystallization engineering [2,73], but reported performance exceeds what operating stability currently licenses. Two physical impediments dominate: interfacial recombination at the hole-selective contact and halide phase segregation in wide-bandgap (WBG) absorbers.

At the buried contact, self-assembled monolayers (SAMs) suffer from solvent-induced degradation and inhomogeneous nucleation. Inserting a 3-TMA molecular interlayer between the self-assembled monolayer (SAM) and the absorber regulates crystallization via hydrogen bonding. That interlayer reaches 23.17% PCE in 1.67 eV and 19.61% in 1.84 eV single-junction devices [2]. Combining graphene oxide with SAMs forms a contact that retains 90% of initial efficiency after 770 h at 25 °C in a 27.3% tandem [74]. Coordinating formamidinium cations with gamma-valerolactone (GVL) cosolvents similarly retards nucleation to enforce oriented micrometer-scale grain growth across the buried interface [73,75].

Within the bulk, mixed-halide absorbers undergo light- and heat-induced phase segregation. Under in-operando thermal cycling from -100 to +100 °C, WBG subcells exhibit severe phase segregation above 50 °C with only partial recovery during dark cooling [76]. Halogen homogenization via TPB-Br surface passivation [77] and conductive polymer grain-boundary bridges [78] suppress halide demixing. These routes yield 23.63% (1.68 eV subcell) and 20.85% single junctions, alongside uncertified tandems of 32.03% on 1.05 cm² and 25.92% [77,78]. Strengthening the metal-halide octahedral framework via Sr²⁺ alloying weakens electron-phonon coupling, achieving 31.66% PCE with an operational lifetime to 80% of initial efficiency (T80) of 6785 h on fully-textured silicon [58].

Upscaling these stacks exposes severe area-dependent transport penalties. Quantitative hyperspectral luminescence imaging on 64 cm² modules reveals that resistive transport and interconnection losses equal intrinsic material losses [79]. Scalable printing using low-toxicity GVL with dimethyl sulfoxide (DMSO) exceeds 17% in WBG cells and 28% in tandems [75], while ionic liquid stabilization demonstrates certified 27.49% tandems on 12.56 cm² with 23.58% certified top-cell efficiency [5].

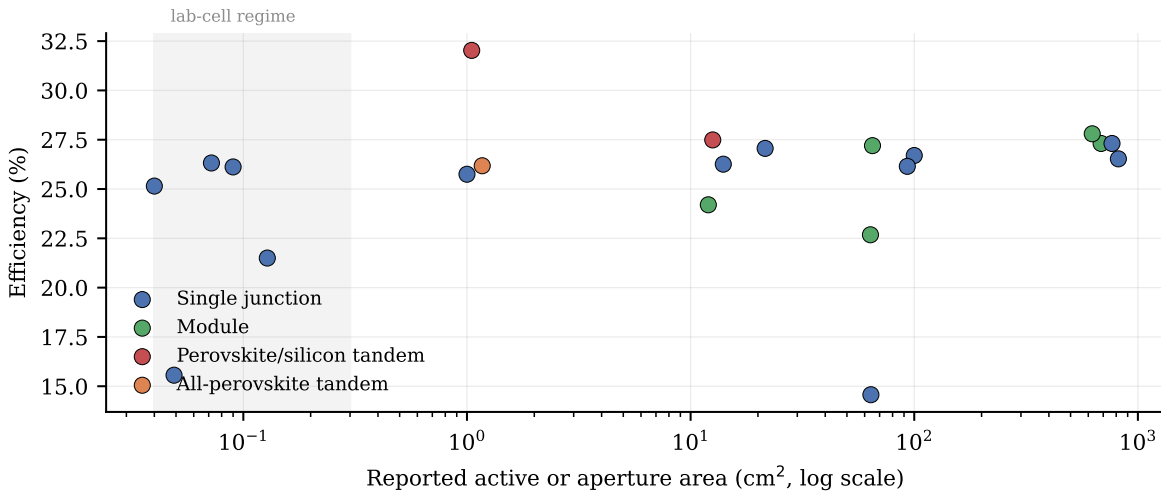


Figure 4. Efficiency against reported active or aperture area. The shaded band marks the sub-0.3 cm² laboratory-cell regime in which most champion values are measured.

8. Research Gaps and Outlook

The verified performance ceiling reaches a certified power conversion efficiency (PCE) of 33.1% in two-terminal (2T) tandems [1] and 27.31% in single junctions [6,7]. Yet the certified reporting share was 5.9% in 2026-06 and 8.3% in 2026-07, with only 22 of 169 closely read papers reporting any certified value. These peak metrics do not license claims of operating durability, as uncertified microscale sweeps mask interfacial degradation, phase instability, and scaling penalties.

First, evidence for contact stabilization fails to establish whether interfacial dipoles and multidentate coordination withstand thermo-mechanical fatigue. While modifying transparent oxides with self-assembled monolayer (SAM) contacts and multidentate chelators elevates initial open-circuit voltage (Voc), thermal expansion mismatches induce cyclic shear stresses that trigger delamination [1,6,35]. Resolving whether coordination bonds survive operating fields requires operando double-cantilever beam fracture testing paired with bias-dependent Kelvin probe force microscopy during thermal cycling.

Second, reported chemical treatments do not demonstrate permanent suppression of halide segregation in wide-bandgap (WBG) absorbers. Although molecular interlayers and B-site alloying achieve high initial efficiency in tandems [2,58], in situ thermal cycling reveals phase segregation above 50 °C with only partial recovery upon cooling [76]. Differentiating true thermodynamic phase stability from kinetic demixing arrest demands operando nanoscale photoluminescence spectroscopy synchronized with diffraction profiling under continuous electrical bias and elevated temperature.

Third, unstandardized testing leaves the boundary between reversible ionic polarization and irreversible decomposition unquantified. Only 4 papers reported a time to 80% of initial performance (T80) lifetime, the longest 9800 h [19], and only 6 papers named an International Summit on Organic Photovoltaic Stability (ISOS) protocol [62,69]. Establishing ion accumulation rates under operating conditions requires operando intensity-modulated photovoltage spectroscopy coupled with secondary-ion mass spectrometry during maximum power point tracking (MPPT).

Fourth, microscale defect passivation fails to establish scalability across large apertures. Only 27 of 648 works addressed scale-up, and the largest reported aperture reached 818 cm² [9]. Passivators that eliminate traps on sub-0.1 cm² apertures aggregate during meniscus drying, intensifying non-radiative recombination across laser-scribed modules [42,79]. Spatially resolved electroluminescence imaging calibrated against local series-resistance maps during coating would isolate interconnect losses. Cross-study evaluation of interfacial stabilization failed because studies reported initial unmasked voltages rather than quantifying interfacial bonding enthalpies alongside continuous operational fill-factor retention under bias.

Acknowledgements

The authors would like to acknowledge the financial support from Xiamen University Malaysia through the Xiamen University Malaysia Research Fund (XMUMRF/2022-C10/IENG/0046), and from National Natural Science Foundation of China (Ref. No.: 52503390).

Declaration of Competing Interest

The authors declare no conflict of interest.

Data Availability Statement

The corpus table for the month, the per-paper extraction records with their verbatim source quotations, the statistics file underlying every number in the text, and the gate report are provided as Supplementary Information. Abstract text and full texts are not redistributed.

AI Usage Declaration

This manuscript was prepared as part of our Research Project for making a reliable Autonomous Researcher Agent. The agent executed the monthly review pipeline without human intervention: it harvested the indexed literature for the month through the OpenAlex, Semantic Scholar, Crossref and arXiv interfaces, applied a deterministic topical filter, ranked and selected the papers read closely, extracted every quantitative claim into a structured record binding each number to a verbatim quotation from that paper’s own abstract, computed the statistics and figures from those records, and drafted this text from the resulting evidence briefs. Literature harvesting, selection, statistics, figures and gate checking are deterministic scripts rather than model output. Structured extraction used qwen3.8-flash through the opencode interface; the prose was written by gemini-3.8-flash-high through the agy interface; independent review used Opus 5 at high reasoning effort. No figure is a model-generated image: every panel is plotted directly from the extracted records. Every quantitative statement in the main text resolves to the machine-generated statistics file shipped as Supplementary Information, and every citation resolves to a record carrying its source quotation. The author reviewed and edited the content, verified the selection, and takes full responsibility for the content of the published article.

References

1. Xu, X. et al. Noncoplanar multidentate contacts reinforce buried interface for perovskite/silicon tandem solar cells. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aei2945>
2. Wang, F. et al. Beyond Self-Assembled Molecules: Hydrogen-Bonding-Regulated Crystallization of Wide-Bandgap Perovskite for High-Performance Perovskite-Silicon Tandem Photovoltaics. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.1676645>
3. Liu, Z. et al. Homogenizing Sn–Pb Distribution Through A-Site MA Cations for Efficient All-Perovskite Tandem Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74078>
4. Lp, H. et al. Enhancing Sn-Pb perovskite homogeneity via thioether coordination for efficient and stable all-perovskite tandem solar cells. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aeb8790>
5. Xiong, Y. et al. Radiation-resilient monolithic wide-bandgap perovskite/p-type heterojunction silicon tandem solar cells for space photovoltaics. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aef6600>
6. Li, B.E. et al. Unlocking 27.3% Perovskite Solar Cells and Ultra-Stable Flexible Modules via Multidentate Molecular-Mediated Buried-Interface Reconstruction. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.73807>
7. Zhang, Z.Y. et al. Unlocking 27.3% perovskite photovoltaics by interface-locked dual-molecule contact. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aeg1456>
8. Gao, D. et al. n-Type polymer layers enable efficient, scalable, and thermally stable perovskite

- solar modules. *Science* (2026). <https://doi.org/10.1126/science.aed8175>
9. Qian, C. et al. Bidirectional Chemo-Mechanical Interface Stabilization in Perovskite Solar Cells. *Journal of the American Chemical Society* (2026). <https://doi.org/10.1021/jacs.6c06932>
 10. Yang, R. et al. Soft supermolecule stabilized buried interface for high-performance inverted perovskite solar cells and modules. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-74018-8>
 11. Shang, X. et al. A Multisite Molecular Salt Enables Surface Passivation and Energetics Management for Efficient Inverted Perovskite Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71305>
 12. Fan, R. et al. Plasma surface engineering for efficient and stable perovskite solar cells and modules. *Science* (2026). <https://doi.org/10.1126/science.aeg1730>
 13. Wong, C. et al. Rational Tailoring of Hole-Selective Self-Assembly Monolayers Based on Sulfur-Containing Heterocycles for High-Performance Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.1864800>
 14. Zhang, H. et al. Dual-Site Interface Passivation by Sulfur-Based Small Molecules for the Simultaneous Enhancement of Efficiency and Stability in Inverted Perovskite Solar Cells. *ACS Applied Materials & Interfaces* (2026). <https://doi.org/10.1021/acsami.6c05331>
 15. Tan, Y. et al. Synergistic Interfacial Dangling-Bond Passivation for Enhanced Moisture Resistance in Perovskite Photovoltaics. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.73950>
 16. Saeed, A. et al. Amphoteric Anion-Mediated Crystallization and Universal Defect Passivation for High-Efficiency and Stable Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77378>
 17. Chen, R. et al. Reconstructing the perovskite surface with a homogeneous electric potential by reducing localized charge for efficient and durable perovskite solar cells. *Energy Materials* (2026). <https://doi.org/10.20517/energymater.2026.102>
 18. Jiang, X. et al. Three-dimensional distribution of amino groups improves perovskite crystallization and defect passivation in high-performance photovoltaics. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-75990-x>
 19. Zheng, W. et al. Intermediate-Site Anchoring Ligands Enable Robust Nonlayered Interfacial Passivation for Efficient and Stable Air-Processed Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74243>
 20. Wu, J. et al. Aniline-Fluoroalkyl Interfacial Passivation Enables Defect-Suppressed and Moisture-Resistant Perovskite Solar Cells. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaem.6c01560>
 21. Deng, Z. et al. Precise Iodide-Trap Strategy for Photothermally Stable Spiro-OMeTAD-Based Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.2924886>
 22. Du, X. et al. Enhancing Interfacial Stability of FAPbI₃ Perovskite Solar Cells by Introducing 2D/3D Heterojunction Based on Functional Imidazolium Cations. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77015>
 23. Huang, C. et al. Phase Separation Suppression by Guanidine Phosphate Enables 21.54% Wide-Bandgap Perovskite Cells for All-Perovskite Tandem Solar Cells. *Carbon Neutralization* (2026). <https://doi.org/10.1002/cnl2.70189>
 24. Cheng, J. et al. Modulating Growth Kinetics and Defect Passivation in Ambient-Air Evaporation-Spray Deposition Enables Efficient and Stable Rigid/Flexible Perovskite Solar Cells.

- EcoEnergy* (2026). <https://doi.org/10.1002/ece2.70102>
25. Lv, D. et al. Dual-Site-Binding Ligand-Driven (100) Orientation for Efficient and Stable Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74033>
 26. Fu, J. et al. Multisite Polymer Regulates Crystallization for Low-Strain Perovskite Film. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70435>
 27. Wang, D. et al. Multi-Bonding Molecular Interface Bridges Enable Synergistic Defect Passivation, Strain Relaxation, and Accelerated Electron Extraction in Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.77373>
 28. Yang, J. et al. Dynamic Strain Regulation Via Photoresponsive Fullerenes Enables High-Performance and UV-Robust Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74220>
 29. Li, M. et al. Improved Mechanical Adhesion Enabled by Bilayer Hole Transport Layers in Perovskite Solar Cells. *ACS Materials Letters* (2026). <https://doi.org/10.1021/acsmaterialslett.6c00479>
 30. Wu, H. et al. A Multidentate Coordination Complex-Based Self-Assembled Monolayer Stabilizes Perovskite Interface. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.8080769>
 31. Duan, Y. et al. Bisphosphonate Self-Assembled Molecules Enhance Anchoring Strength and Perovskite Passivation Toward High-Efficiency Perovskite/Silicon Tandem Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.74999>
 32. Wang, J. et al. Suppression of SAM desorption by engineering polyoxometalate interface for inverted perovskite solar cells. *Polyoxometalates* (2026). <https://doi.org/10.26599/pom.2026.9140147>
 33. Qin, Z. et al. Tetrakisphosphorylated phthalocyanine-based self-assembled monolayer stabilizes perovskite photovoltaics. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aee4175>
 34. Chen, X. et al. Molecular Lubricant Mitigates Self-Assembled Monolayer Aggregation in Perovskite Photovoltaics. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.7090517>
 35. Zhou, B. et al. Proton Transfer Complex-Confined Self-Assembled Monolayer for Thermally Stable Inverted Perovskite Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenenergylett.6c01489>
 36. Wan, X. et al. Strengthened Charge-Selective Perovskite/SAM Heterocontact for Efficient and Durable Perovskite Solar Cells via Interfacial Reconfiguration. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.1018855>
 37. Su, H. et al. Synergistic Interface Modulation of Hole Transport for Efficient, Stable Inverted Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.76878>
 38. Lu, D. et al. Co-Assembled Hybrid Interlayer Engineering for Enhanced Upper Interface Stability in Inverted Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.73886>
 39. Chen, X. et al. Perylenecarbazole-Based Polycyclic Aromatic Self-Assembled Monolayers with Tailored Electrostatic Potentials for High-Performance Organic and Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.4031162>
 40. Ma, P. et al. Ternary Cooperative Interface With Electronegative Phosphine Bridges Enables Efficient and Stable Inverted Perovskite Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74416>

41. Wang, Y. et al. PFN-Br Modified Buried Interface Enhanced Charge Extraction in Inverted Wide-Bandgap Perovskite Solar Cells. *Small* (2026). <https://doi.org/10.1002/sml.74806>
42. Miao, Z. et al. A unified ligand-dimensional design to halt cation migration in perovskite photovoltaics. *Science Advances* (2026). <https://doi.org/10.1126/sciadv.aed6327>
43. LIU, M. et al. In-Situ Constructed Cations for 2D/3D Perovskite Heterostructure for Stable and Efficient Photovoltaics. *Advanced Science* (2026). <https://doi.org/10.1002/advs.76508>
44. Forte, S. et al. Multilayer Assembly of Perovskite Heterostructures via Colloidal Suspensions. *ChemRxiv* (2026). <https://doi.org/10.26434/chemrxiv.15006632/v1>
45. Moazzezi, P. et al. Solvent Esterification and Stoichiometric Control in Ambient-Grown FAPbI₃ Single-Crystal Solar Cells. *Journal of the American Chemical Society* (2026). <https://doi.org/10.1021/jacs.6c05502>
46. Zhang, Q. et al. Sulfolane-Based Solvent Engineering for High-Performance MAPbI₃ Perovskite Solar Cells. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaem.6c01764>
47. Chang, Q. et al. Dual-Function Molecular Modulator Enables High-Density Nucleation and Ordered Crystal Growth for Efficient Tin Perovskite Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.76731>
48. Luo, Y. et al. Impact of Perovskite Surface Termination Dependent Molecular Interaction on Interfacial Energetics and Phase Stability. *ACS Nano* (2026). <https://doi.org/10.1021/acsnano.6c08922>
49. Wang, S. et al. Simultaneous Enhancement of Efficiency and Photostability in Wide-Bandgap Perovskite Solar Cells via Managing Residual PbI₂ with a Phosphoramidate Additive. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaem.6c01370>
50. Zhu, W. et al. Thermodynamically Matched Molten-Salt-Assisted Phase Engineering of 2D/3D Perovskite Heterostructures for Efficient and Stable Solar Cells. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.76974>
51. Jiang, Y. et al. Structure Optimization Strategy of PbI₂ Vacuum Deposition Process for Efficiency Enhancement of Perovskite Solar Cells. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70430>
52. Ryu, J. et al. Resolving Nanoscale Heterogeneities in Lead Halide Perovskites Through Low-Dose Concurrent 4D-STEM-EDX Mapping. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74382>
53. Maziviero, F.V. et al. Computational and experimental evaluation of bismuth modification in CsPbX₃ (X = I— or Br—): Impact on structural and optical properties. *Solar Energy Materials and Solar Cells* (2026). <https://doi.org/10.1016/j.solmat.2026.114597>
54. Zhou, Y. et al. Preventing Triple Halide Intermediate Formation for Segregation-Resistant Wide Band Gap Perovskites. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenenergylett.6c01113>
55. Maksudov, T. et al. Achieving 3.31 V in Triple-Junction Perovskite–Organic Tandem Solar Cells via Lattice Distortion–Induced Phase Stability. *Advanced Functional Materials* (2026). <https://doi.org/10.1002/adfm.76715>
56. Tan, Q. et al. Phenomenological Prediction of Halide Perovskite Phase Stability Phenomenological Prediction of Halide Perovskite Phase Stability via Dielectric-Response Analysis: Equivalence to First-Principles Thermodynamics at Five Orders of Magnitude Reduced Computational Cost. *ChemRxiv* (2026). <https://doi.org/10.26434/chemrxiv.15005541/v1>
57. Wang, Y. et al. Interfacial Hydrogen Bonding for Efficient and Robust Flexible Tin Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.7>

58. Liu, P. et al. Sr/Pb Alloying Stabilizes Wide-Bandgap Perovskites for Durable Fully-Textured Perovskite/Silicon Tandem Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71317>
59. Ni, Q. et al. Protonated Cyclohexylamine as an Effective Passivation Layer for High-Performance Perovskite Solar Cells. *ACS Applied Energy Materials* (2026). <https://doi.org/10.1021/acsaem.6c01637>
60. Ahmad, N. et al. Synergistic Interface and Band Engineering for Sn-Pb Perovskites. *Small* (2026). <https://doi.org/10.1002/sml.74531>
61. Zhou, X. et al. Homogenizing Cesium Distribution via Rubidium Incorporation Enables Pure-Iodide 1.67 eV Bandgap Perovskite Solar Cells with Efficiency Exceeding 22%. *ACS Nano* (2026). <https://doi.org/10.1021/acsnano.6c04084>
62. Huang, J. et al. Multidentate Fullerene Boosts the Thermal Stability of Perovskite Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acseenergylett.6c01517>
63. Chalkias, D.A. et al. Depth-Resolved Photo-Degradation of Perovskite Layer in Solar Cells Stressed Under ISOS Light-Aging Protocol Conditions: The Role of Substrate, Electron-Transport-Layer and Perovskite Deposition Method. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70421>
64. Zhong, S. et al. Molecular Tailoring of Self-Assembled Monolayers Derived From Asymmetric Benzothienoinole for Efficient and Stable Inverted Perovskite Solar Cells and Modules. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.1678852>
65. Yang, G. et al. Central Core Engineering of Aromatic Carbonyl Molecules Enables Highly Efficient and Stable Perovskite Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.5216345>
66. Dai, K. et al. Thermally Independent Interfacial Reconstruction for Phase-Pure n = 1 2D Perovskites With Mixed-Orientation Surface Architecture. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74247>
67. You, S. et al. Boosting Carrier Transport in Carbon-Based Hole-Transport-Layer-Free Perovskite Solar Cells via 2D/3D Interfacial Structural Configuration. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70447>
68. Zhao, J. et al. Eliminating Residue-Induced Degradation: A Volatile Dopant Strategy for High Performance Perovskite Photovoltaics. *Small* (2026). <https://doi.org/10.1002/sml.74289>
69. Kim, S. et al. Maximum Stability Point Tracking Stabilizes Wide-Bandgap Mixed-Halide Perovskite Solar Cells. *Advanced Energy Materials* (2026). <https://doi.org/10.1002/aenm.71358>
70. Kong, W. et al. Stabilizing 1.93-eV ultrawide-bandgap perovskites for efficient triple-junction solar cells. *Nature Communications* (2026). <https://doi.org/10.1038/s41467-026-75305-0>
71. Fasti, R. et al. Dynamic Stress and Strain Evolution and Engineering Strategies to Enhance Operational Durability in Perovskite Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acseenergylett.6c01837>
72. Guo, B. et al. A Composition-Dependent Stress-Diffusion Framework Identifies Humidity-Tolerant Processing Windows in Hybrid Perovskite Solar Cells. *ChemRxiv* (2026). <https://doi.org/10.26434/chemrxiv-2025-128tn/v2>
73. Liu, L. et al. Artificial Intelligence-Guided Cosolvent Design for High-Performance Perovskite/Silicon Tandem Solar Cells. *Nano-Micro Letters* (2026). <https://doi.org/10.1007/s40820-026-02291-9>
74. Yun, Y. et al. Triple-junction all-perovskite solar cells with self-assembling hole contacts in all

- subcells. *Joule* (2026). <https://doi.org/10.1016/j.joule.2026.102575>
75. Danguddubiyam, U.K. et al. Eco-Friendly Solvent System for Inkjet Deposition of Wide Bandgap Perovskite Solar Cells Enabling Tandem Integration. *Advanced Science* (2026). <https://doi.org/10.1002/advs.76463>
 76. Durant, B.K. et al. In-Operando Thermal Cycling of All-Perovskite Tandem Solar Cells. *ACS Energy Letters* (2026). <https://doi.org/10.1021/acsenergylett.6c00992>
 77. Chen, Y. et al. Molecular-Extrusion-Driven Halogen Homogenization for Efficient Perovskite-Silicon Tandem Solar Cells. *Angewandte Chemie International Edition* (2026). <https://doi.org/10.1002/anie.5530609>
 78. Zhou, Y. et al. Conductive Polymer Bridges Mediate High-Performance Flexible Perovskite–Organic Tandem Solar Cells. *Advanced Materials* (2026). <https://doi.org/10.1002/adma.74183>
 79. Harada, N. et al. Quantitative Hyperspectral Luminescence Imaging Reveals Upscaling Losses in Perovskite Solar Modules. *Solar RRL* (2026). <https://doi.org/10.1002/solr.70375>